Direct imaging of extra-solar planets

J.-L. Beuzit
LAOG, Grenoble

Limitations of indirect detections

- Radial Velocity
  - Detection sensitive to orbit inclination
  - Indetermination on the planet mass
  - Studies limited to Solar type stars or cool stars. Preferably non active stars
  - Need for long observing periods
- Photometric transits
  - Very sensitive to inclination
  - Close planets, large radii

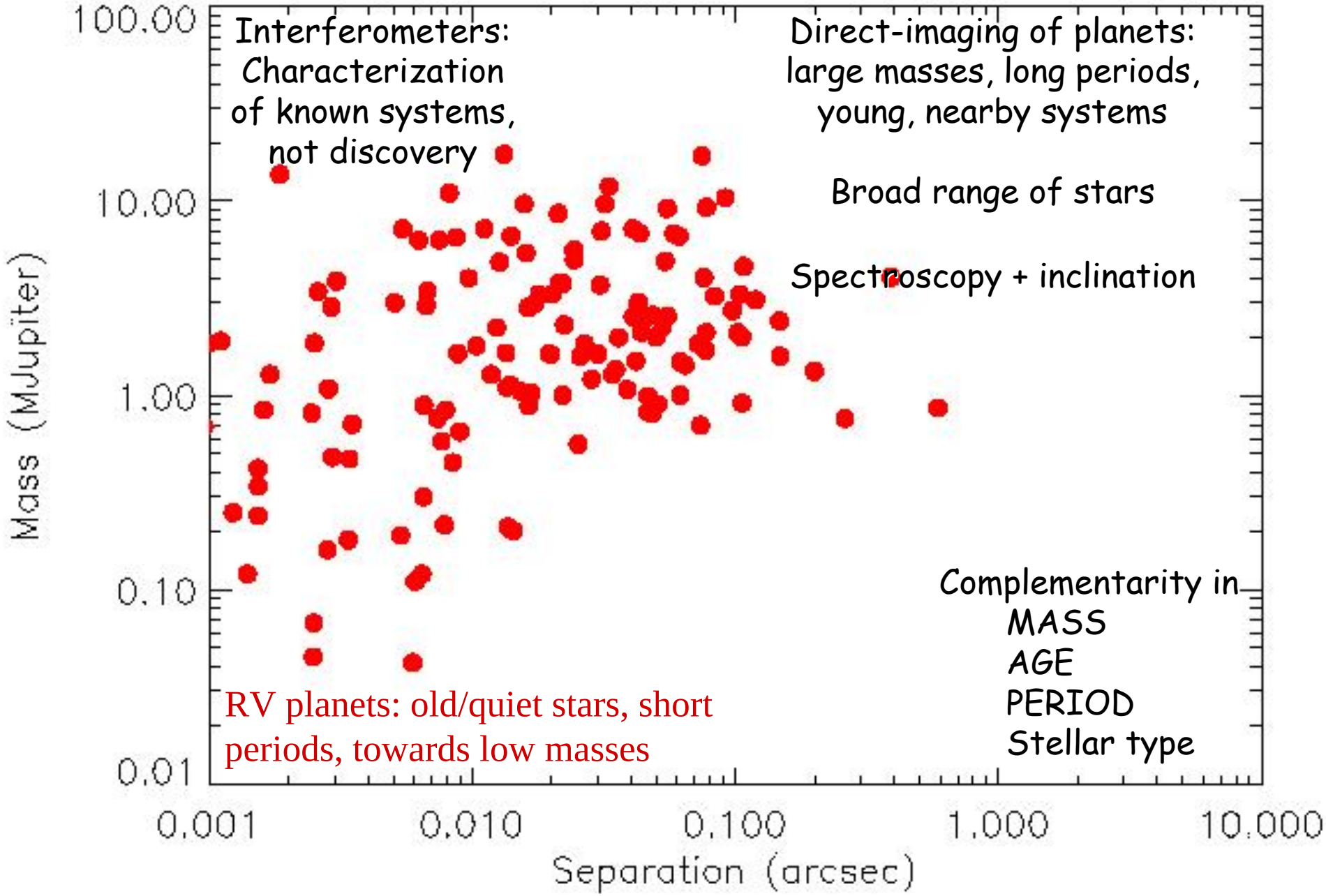
SPHERE

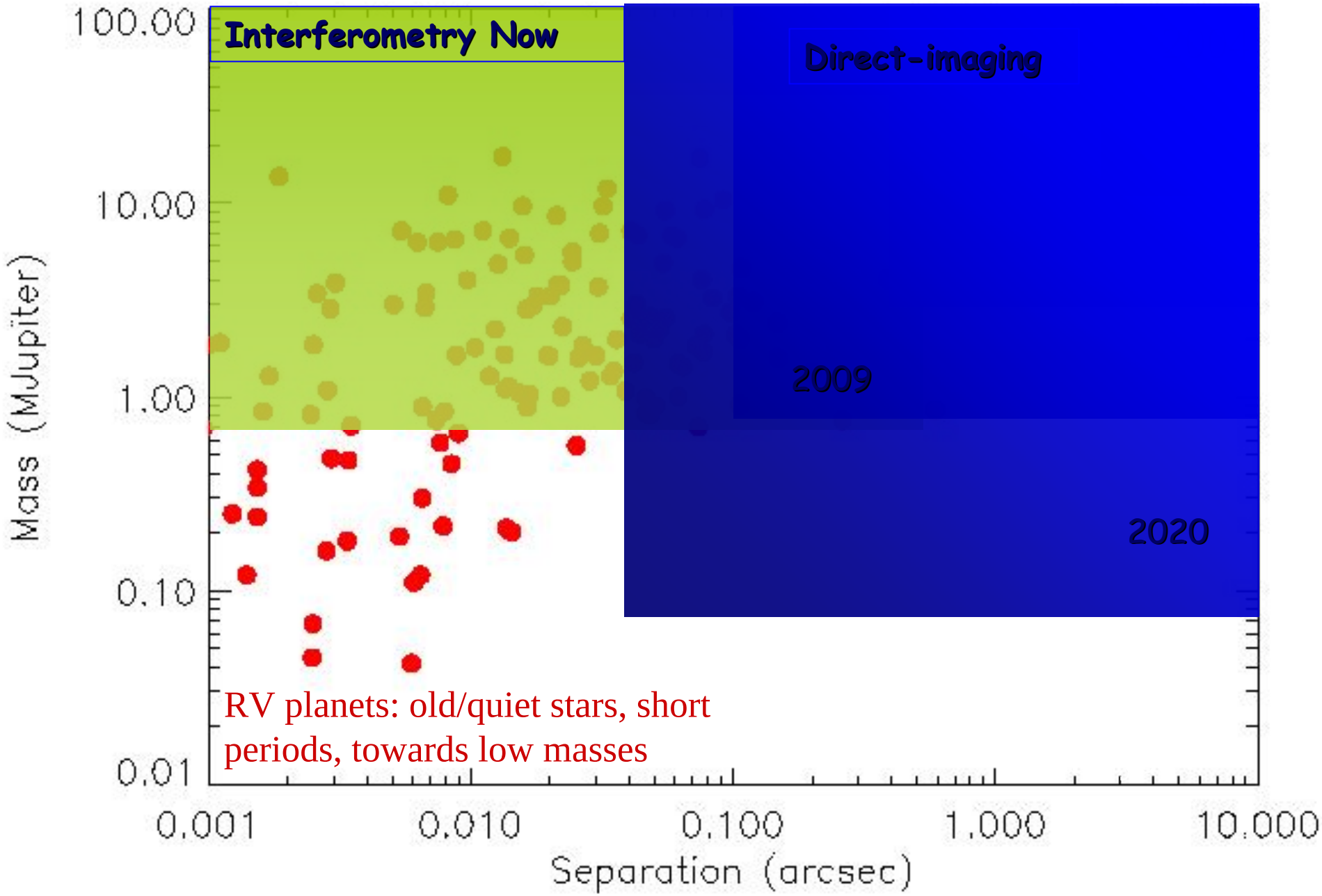
Imaging of extra-solar planets

- Rapid identification of the planet
- Determination of the orbit, no M.*sin(i)* ambiguity
- Characterization of the planet
  - albedo, temperature, chemical composition

  => test of atmospheric models, of evolutionary models
- Access to a new (separation, age) domain
  - Planets around young stars
  - Larger separations (P > few years)
- Access to *all* types of stars
  - Early type stars
  - Active stars

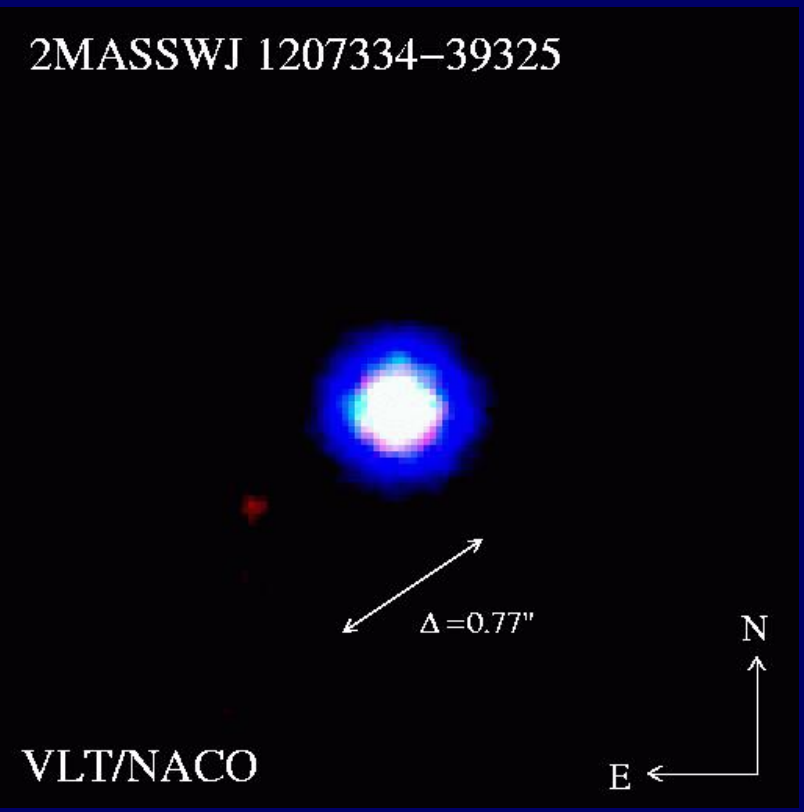
But … difficult !

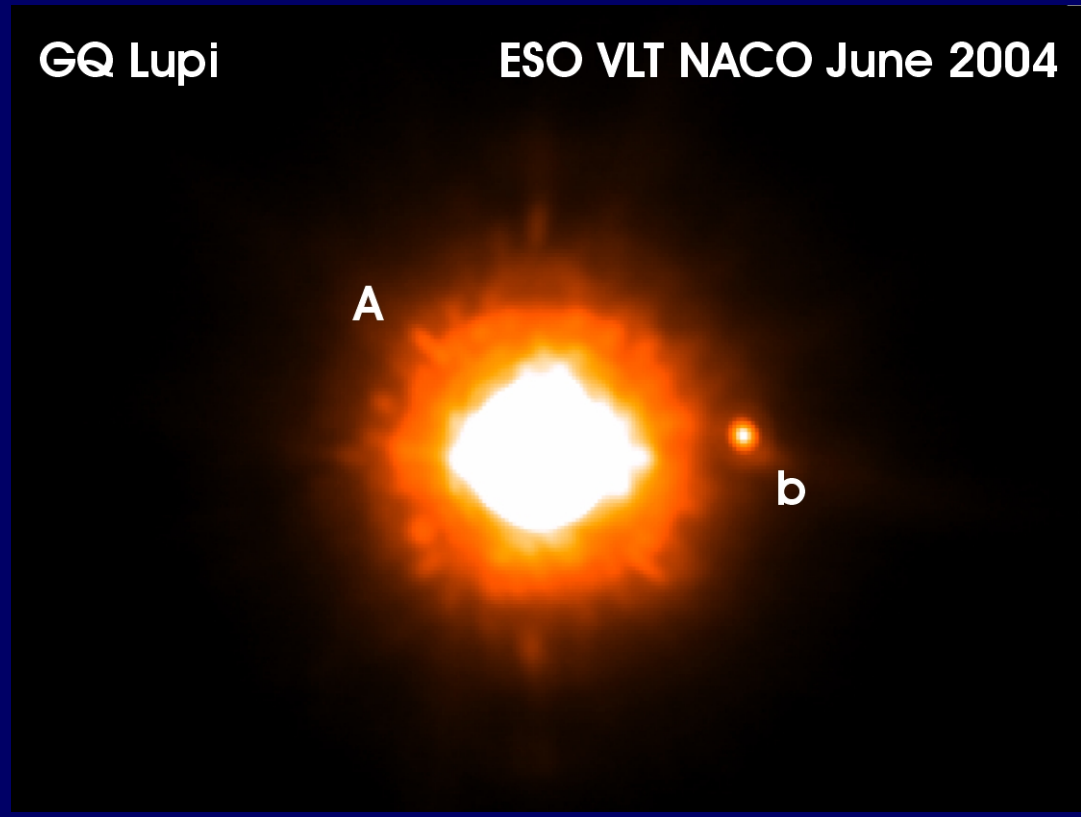

Interferometers:
Characterization
of known systems,
not discovery
Direct-imaging of planets:
large masses, long periods,
young, nearby systems
Broad range of stars
Spectroscopy + inclination
Complementarity in
MASS
AGE
PERIOD
Stellar type
RV planets: old/quiet stars, short
periods, towards low masses
100.00
10.00
1.00
0.10
0.01
Mass (MJupiter)
0.001
0.010
0.100
1.000
10.000
Separation (arcsec)


Interferometry Now
Direct-imaging
2009
2020
RV planets: old/quiet stars, short periods, towards low masses
100.00
10.00
1.00
0.10
0.01
Mass (MJupiter)
0.001
0.010
0.100
1.000
10.000
Separation (arcsec)

SPHERE

And we have results now!





Extremely young ~M_J planets

SPHERE

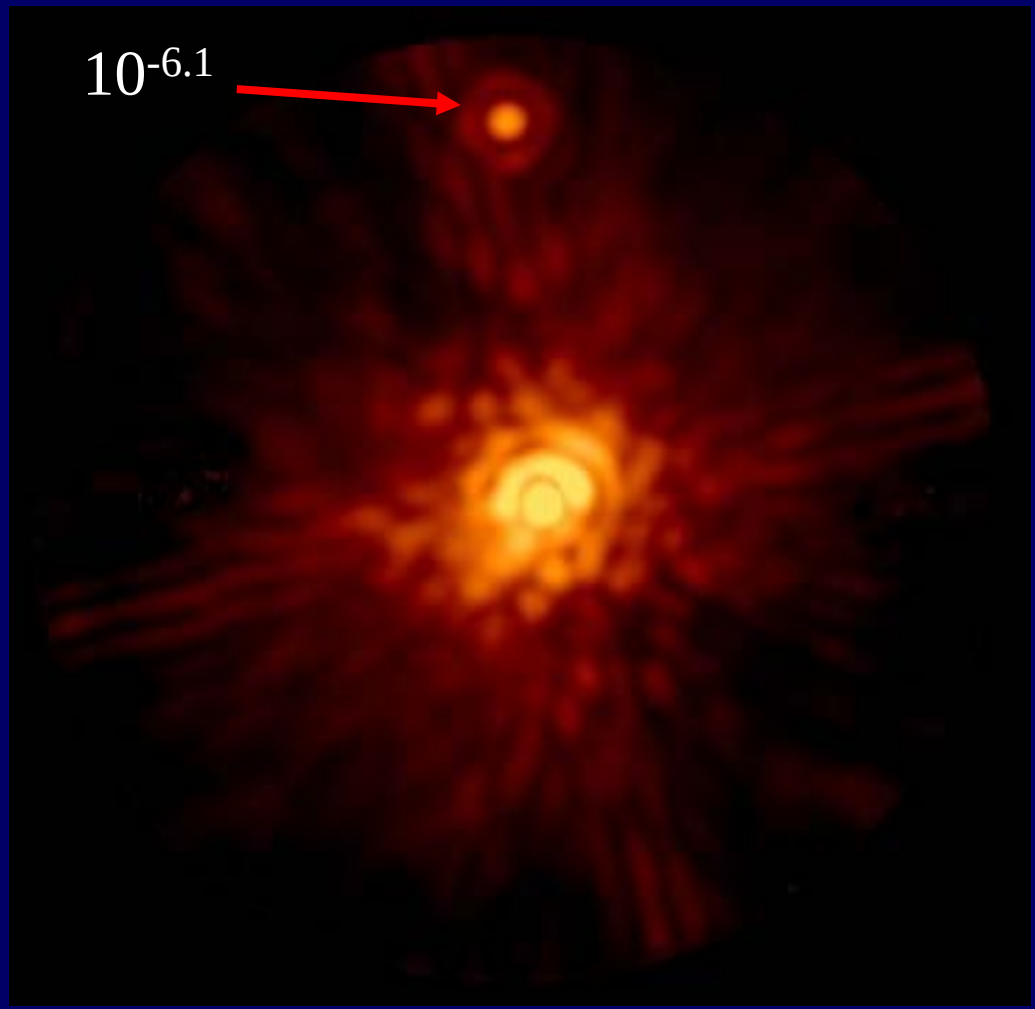
Some Nearby Star

H-Band
15 minute exp.

Weird colors
(I, z, J, H, K)

Companionship
not confirmed

Courtesy
B. Oppenheimer
LYOT project



Requirements:
EGP: $10^{-6}/10^{-8}$
Earth: $10^{-9}/10^{-10}$

Facilities:
VLT-ELT + XAO
Space

5”